

Research Activities of Prof. Longhai Li

Overview of Prof. Li's Research Activities

Prof. Li's research focuses on developing and applying statistical machine-learning methods to analyze bioinformatics and epidemiology data. His research activities are computationally intensive, aiming to develop new data analysis software and solve real problems in life and health sciences.

Predictive Methods for Model Validation

He aims to develop new tools for evaluating Bayesian/non-Bayesian models with complex structures. Today, increasingly complicated models are being proposed for a variety of correlated data such as temporal, spatial, and repeated measurement data. There is a gap between developing new modelling methods and model validation methods. He is working on developing new residual diagnostic methods for checking the adequacy of statistical models.

Statistical Machine Learning

He aims to develop new tools for honestly measuring the predictivity (such as error rate, AUC) of selected features, and new tools for identifying truly predictive features and for building sharper predictive models for phenotypes. He is particularly interested in uncovering the molecular mechanisms behind Alzheimer's and Parkinson's diseases.

Funded Research Projects

Many granting agencies including NSERC, CFI, CANSSI, CFREF, and MITACS have supported his research; see [his research funding history](#).

Past and Current Team Members

- [Post-doctoral Fellows](#)
- [Graduate Students](#)
- [Undergraduate Students](#)

Publications

- [Papers in Referred Journals](#)
- [Software Released Publicly](#)
- [Online Apps/Websites](#)
- [Referred Conference Publications](#)
- [Presentations](#)
- [Preprints](#)

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